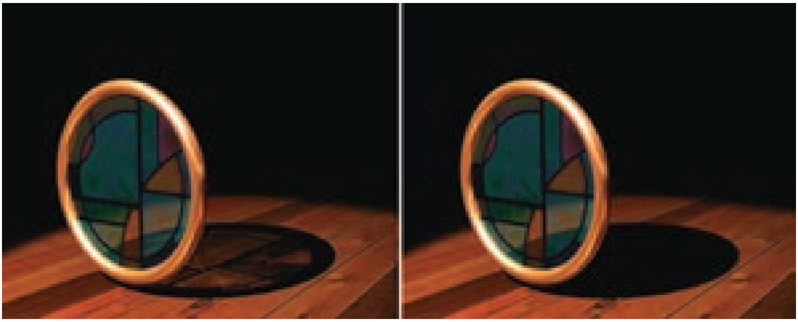

Raytraced shadows (left) render shadows through transparent objects, but depth map shadows (right) won't.

softness of the shadow, along with controls to optimize transparency and shadow calculation.

4.4 Creating Lighting Effects

Lighting Effects add realism to the way the light illuminates the scene. Adding decay and attenuation simulates the way a real-world light's energy falls off with distance. These parameters can also be used to play with reality and place or remove light from the scene.

Decay

By default, lights in 3ds Max illuminate all objects equally, no matter how far they are from the light. In the real world, however, the intensity of a light decays with distance. To create more natural lighting that falls off over distance, you can select Decay in the light's Intensity/ Colour/ Attenuation rollout. There are two types of decay:

- Inverse falls off directly with distance.
- Inverse Square falls off as the square of the distance. This is the same as real-world lighting.

When decay is added to a light, you almost always need to increase the intensity of the light to illuminate the scene. When using Inverse Square decay, you may have to increase the light intensity by several orders of magnitude to illuminate the scene. One way to offset this is to use the Start parameter, which specifies a discrete distance from the light where the decay starts.

Attenuation

Another way to control the way light changes over distance is to use near and